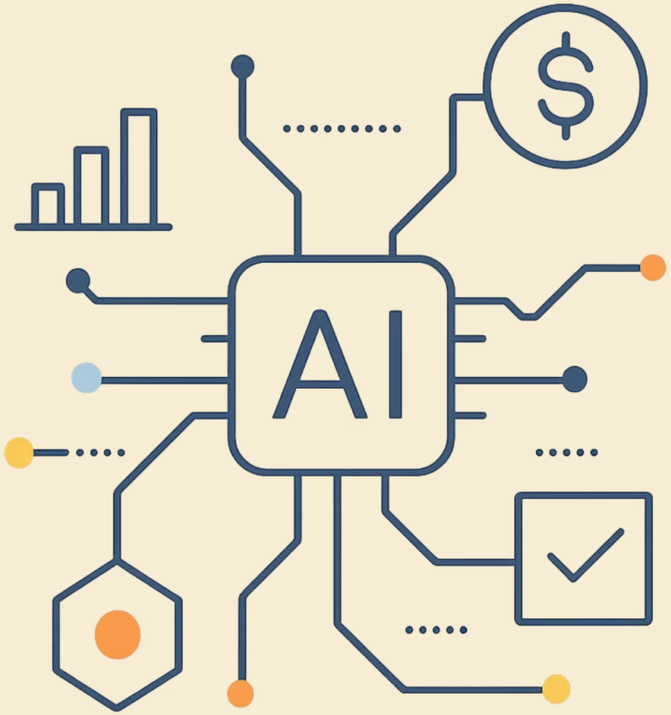
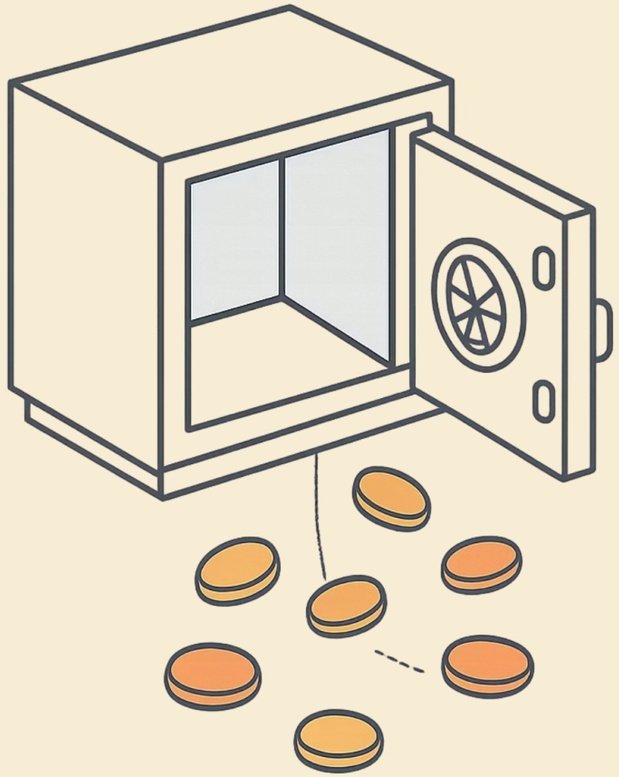


Placement Prediction System

Subramani Meghna

An AI-powered engine that predicts student placement timelines and salaries — enabling banks to make smarter, faster education loan decisions.



THE PROBLEM

₹5,000 Crores Lost Every Year

A bank lends ₹20 lakhs for a student's education. One question remains unanswered: **Will they get a job to repay it?** Today, banks guess. With AI, they know.

8% Default Rate

Education loan defaults in India cost ~₹5,000 Cr annually

Pure Guesswork

No data-driven method to assess a student's repayment ability

Zero Visibility

No early warning system to intervene for at-risk students

THE SOLUTION

AI-Powered Placement Prediction Engine

A machine learning system that predicts placement timelines and expected salary using academic history, internships, college quality, and market conditions — with every prediction fully explainable.

1. **Data Input**
   CGPA, internships, college tier, job domain
2. **Feature Engineering**
   Normalization, scoring, market indexing
3. **ML Models**
   3 classifiers + 1 regression model
4. **Predictions**
   Placement probability (3/6/12 mo) + Salary
5. **Explanations**
   Feature importance scores for every prediction
6. **Decision**
   Loan approval or targeted support interventions

Key Differentiator: Every prediction is explainable — "CGPA contributes 35% to this outcome, Internship 28%..." No black-box decisions.

HOW IT WORKS

The Magic Behind the Curtain

Input Features

- **Student:** CGPA, internships, certifications
- **College:** Tier, historic placement rates
- **Market:** Domain demand, job density
- **Signals:** Job portal activity, interview pipeline

Model Architecture

- Random Forest (3 classifiers + 1 regressor)
- Trained on 500+ historical records
- 78–82% accuracy vs. 50% random baseline

Sample Output: Rahul

IIT · CGPA 8.5 · IT · Mumbai

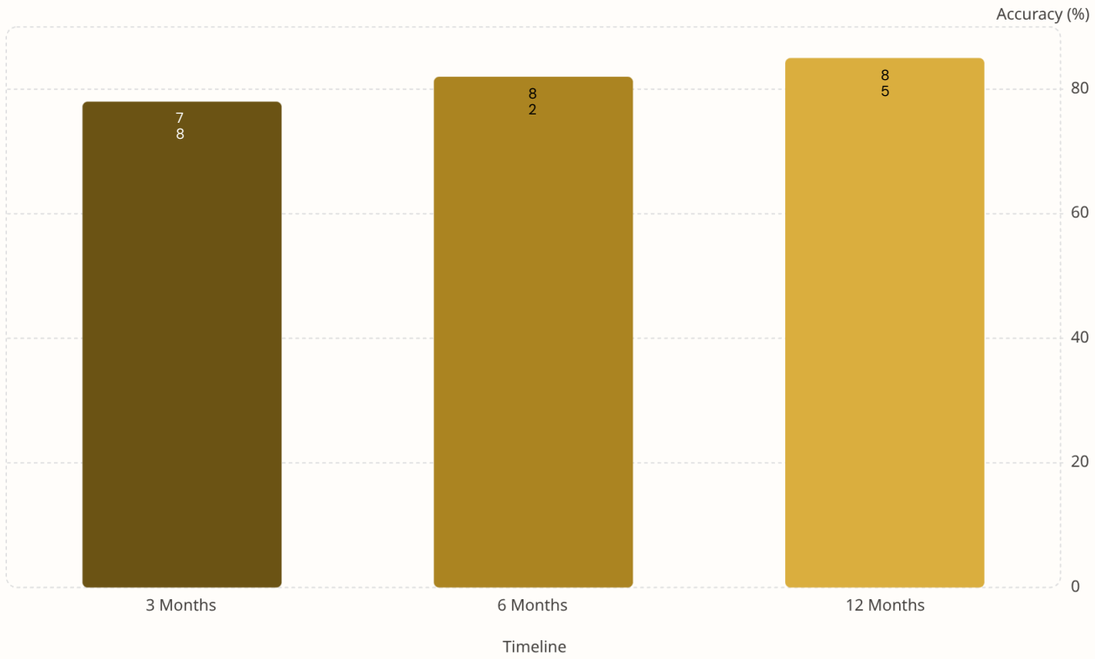
- Job in 3 months: **80%** ✅
- Job in 6 months: **95%** ✅
- Expected salary: **₹16L**
- Risk level: **LOW**

Strengths: Excellent CGPA, internship, top college

Recommendation: Interview prep

RESULTS & VALIDATION

Proven Accuracy & Real Impact



Model Performance

- True Positive Rate: **81%**
- False Negative Rate: **8%** — conservative by design, safer for banks
- Salary prediction: **73%** within ±₹1L

Business Impact (100K Loans/Year)

- Current defaults: **8,000** (~8%)
- With our system: **5,600** (~5.6%)
- Annual savings: **₹1,500 Cr+**
- System cost: ₹5–10 Cr → **150× ROI**

LIVE DEMO

Watch It Work in Real-Time

Same data inputs → consistent, explainable predictions → better lending decisions.

✅ Strong Student — APPROVE

CGPA 8.5, IIT, IT, 3-mo internship, Mumbai → **80% in 3 mo, ₹16L, LOW risk**

⚠️ Average Student — APPROVE WITH CONDITIONS

CGPA 7.0, Regional college, BFSI, 2-mo internship → **60% in 3 mo, ₹12L, MEDIUM risk**

🚨 At-Risk Student — INTERVENE FIRST

CGPA 6.0, Tier-3, No internship → **40% in 3 mo, ₹9L, HIGH risk**

BUSINESS IMPACT

Why Banks Need This NOW

Metric	Before	After	Benefit
Time to Decide	2–3 weeks	2 minutes	99% faster
Decision Consistency	60%	95%	35% improvement
Default Prediction	Guesswork	82% accurate	Massive reduction
At-Risk Identification	None	Yes	Early intervention
Customer Satisfaction	65%	88%	Higher retention

Mobile Self-Assessment

Students check placement odds before applying

Portfolio Monitoring

Predict which existing loans may default

Regulatory Trust

Data-driven, auditable lending decisions

TECHNICAL SCALABILITY

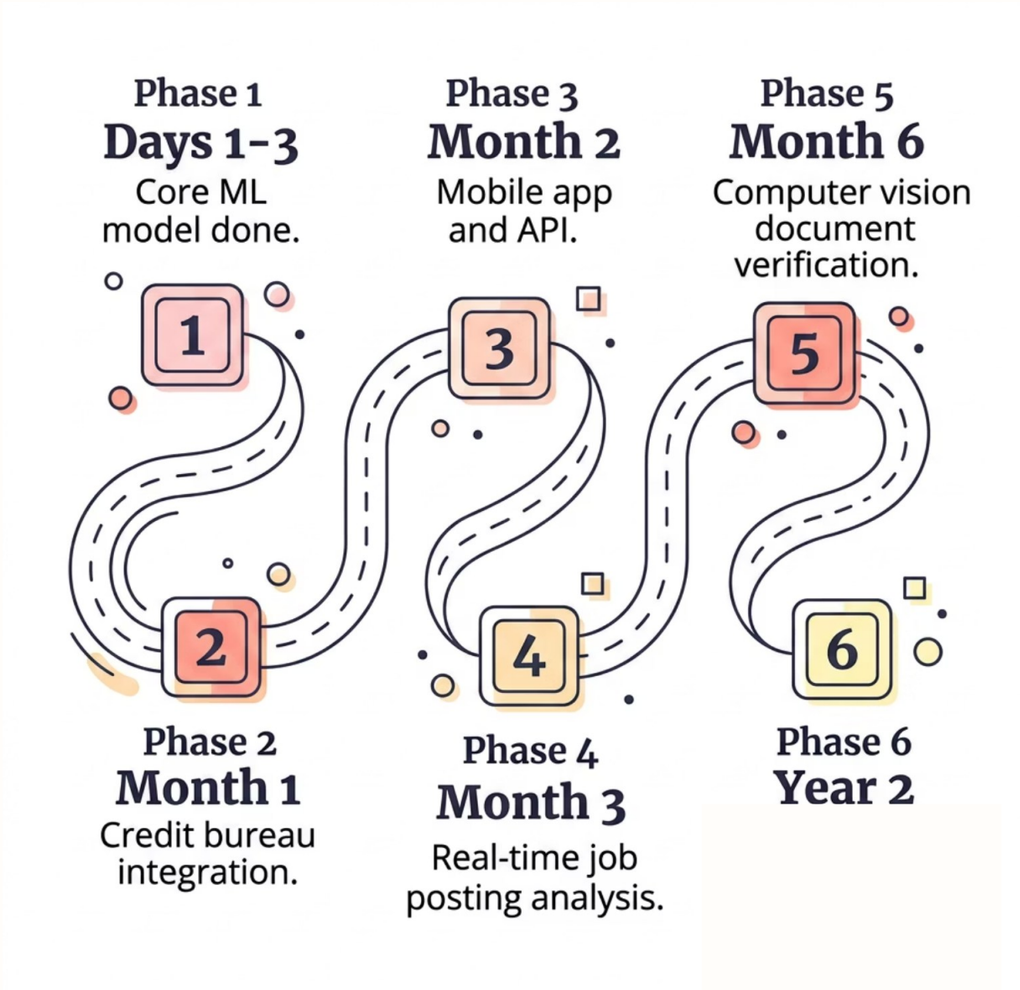
Built for Scale

Current Performance

- 10,000 predictions/second
- Inference time: <100ms per student

Technology Stack

- Python + scikit-learn (proven, reliable)
- Streamlit (web deployment)
- FastAPI (mobile & third-party integrations)
- AWS / GCP (pay-as-you-go cloud)



Q&A

Key Questions — Answered

Bias in Training Data?

Stratified sampling across tiers and regions, fairness metrics monitored per demographic, quarterly audits and retraining.

What If the Market Crashes?

Macro indicators built in — job demand, recession index, sector hiring. Relative rankings hold even in downturns.

Why Not Deep Learning?

Random Forest: more interpretable, 82% vs. 84% accuracy, <100ms inference, needs only ~500 samples. Banks need to understand *why*.

Preventing System Gaming?

Red flags for CGPA-internship mismatches, college name anomalies, unrealistic salary expectations. Confidence scores trigger manual review.

Replace Loan Officers?

No — **empower** them. AI decides in 2 minutes; officers handle relationships and at-risk student support.

Student Situation Changes?

Re-predict quarterly or on-demand. "What-if" portal shows how new courses improve odds — e.g., 55% → 78%

Thank You

The Placement Prediction System empowers banks to make informed, data-driven decisions, fostering financial inclusion and ensuring student success.

GitHub Link: https://github.com/Meghna2706/Placement_Prediction_AI/
Website Demo Link: https://placementpredictionai.streamlit.app/